

STUDENT SKILL-BASED PROJECT RECOMMENDATION SYSTEM

A Literature Survey and Comparative Review

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Abstract— Selecting a suitable academic project is an important task because students have different levels of technical knowledge, interests, programming experience, and career goals. Traditional project selection often depends on manual choice, preference lists, or faculty recommendations. Such methods may not always match students with projects that fit their existing skills. This paper presents a Student Skill-Based Project Recommendation System that recommends academic projects by matching student profiles with project requirements. The proposed system maintains student interests, programming languages, technical skills, academic information, and career preferences. Project records contain required skills, domain, technologies, and difficulty level. A recommendation module compares these attributes and produces ranked project suggestions. Faculty members can review the recommendations before final selection. The system can further support progress monitoring, feedback, evaluation, and reporting. The main objective is to improve project suitability, personalization, student satisfaction, and academic outcomes while reducing manual effort.

Keywords— Student skills, project recommendation, skill matching, academic project, personalized recommendation, technical skills, recommendation system, machine learning, student interests, project allocation

I. INTRODUCTION

Academic project selection is a common challenge in higher education institutions. Students are expected to choose a project that is technically achievable, academically relevant, and useful for their future career. However, students may have very different strengths in programming, databases, web development, networking, artificial intelligence, data science, cloud computing, and software engineering. Projects also have different technology requirements and levels of complexity.

A Student Skill-Based Project Recommendation System addresses this challenge by treating project selection as a matching and recommendation problem. Instead of depending only on student preference, the system considers the student's actual technical skills and interests together with the requirements of each project. The outcome is a ranked list of projects that are more closely aligned with the student's capabilities.

The proposed approach is useful because it can support students before project allocation, help faculty members guide project selection, and provide a structured basis for future academic analysis. The system can also be extended to use historical project outcomes so that recommendations become more accurate over time.

The remainder of this paper describes the review methodology, literature survey, comparative analysis, limitations, research gaps, proposed system, future directions, and conclusion.

Additional Discussion

The proposed recommendation approach also improves transparency because the student can see which skills contributed to the recommendation. For example, a project can be shown with matched skills such as Java, SQL, HTML and Python, together with the percentage of required skills already available. This gives students a clear reason for selecting one project over another and also helps faculty members discuss the recommendation with the student. In addition, the system can identify projects that are slightly above the current skill level and present the missing skills as learning requirements.

Further Discussion

The system can make project selection more systematic by presenting a ranked list rather than forcing students to search through every available project. Each recommendation can include the matched skills, interest area, difficulty level and the reason for its ranking. This improves clarity and gives students a practical basis for discussion with faculty.

A hybrid approach can combine technical skill matching with interest matching and academic guidance. Technical skills represent readiness, interests represent motivation, and faculty review represents academic judgement. Combining these factors can provide a balanced approach that is practical for a college environment.

Security and role-based access are important because student academic information must be visible only to authorized users. Students can manage their own profiles, faculty can view assigned students, and coordinators can maintain project information. Authentication and access control should therefore be included in the implementation.

Implementation Note

The proposed system is designed to remain practical for academic use. It can start with a simple matching mechanism and gradually incorporate more advanced recommendation techniques. The important requirement is that the student receives understandable, relevant project suggestions based on current skills and interests, while faculty members retain the ability to review and approve the final choice.

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II. REVIEW METHODOLOGY

The literature review was organized around student-project allocation, preference-based matching, skill-based recommendation, recommender systems, machine learning, faculty guidance, and academic project management. Relevant studies were considered based on their relationship to student-project matching and recommendation. For each research direction, the problem addressed, methodology, data or tools, findings, and limitations were examined.

The review particularly focuses on the difference between traditional allocation and personalized recommendation. Traditional allocation mainly answers the question of which student receives which project. A recommendation system goes one step further by estimating which projects are most suitable for a specific student. This distinction is important when project requirements and student capabilities vary considerably.

A. Student-Project Allocation

Student-project allocation research commonly models students, projects, preferences, and capacity constraints. Stable matching provides a formal approach for producing allocations while respecting preferences and project or lecturer limits. These studies establish an important foundation for automated assignment.

B. Preference-Based Approaches

Preference-based systems allow students to rank projects according to their interests. Preference information can improve satisfaction and can be useful in allocation algorithms. However, preference alone cannot guarantee that a student has the technical skills needed to complete a selected project.

C. Skill-Based Matching

Skill-based matching introduces technical capabilities into the recommendation process. A student may have skills in Java, Python, SQL, HTML/CSS, machine learning, cloud services, or networking. Projects can be represented with corresponding required skills. Matching the two sets provides a direct measure of suitability.

D. Personalized Recommendation

Personalized recommendation systems generate different results for different users. In the academic project context, two students with different skill profiles should not necessarily receive the same project ranking. Personalization can therefore make project selection more meaningful and practical.

E. Machine Learning

Machine-learning methods can be introduced when enough historical data is available. Student profiles, selected projects, project completion, evaluation marks, and feedback can be used to learn patterns of suitability. However, model quality depends on reliable and sufficiently large data.

F. Role of Faculty

Automated recommendations should support rather than replace academic guidance. Faculty members can review suggested projects, check technical feasibility, consider team composition, and provide final guidance before project confirmation.

G. Project Lifecycle

A complete academic platform can connect recommendation with milestones, weekly progress, deadlines, feedback, evaluation, and reports. Such integration allows the institution to study whether suitable recommendations are associated with better project outcomes.

Additional Discussion

The recommendation process can be updated whenever the student changes the profile. New programming languages, certifications, completed courses and project experiences can increase the student's matching score for relevant projects. This dynamic profile makes the recommendation more useful than a one-time allocation process. It also supports continuous academic guidance because faculty members can observe how the student's interests and technical abilities develop over time.

Further Discussion

Student profiles should be kept current because technical abilities change throughout a degree program. New programming languages, tools, certifications, internships, laboratory work and completed projects can be added to the profile. Updating these details allows the recommendation engine to generate more relevant project suggestions.

The first implementation can use a simple weighted score because it is transparent and easy to explain. Later, historical project data can be used to evaluate machine-learning methods. Any advanced model should be compared with the simple baseline using measurable evaluation parameters such as recommendation relevance, student satisfaction and project completion.

In future, the system can be evaluated using real student and project data. The results can be used to improve recommendation rules, discover skill gaps, and understand whether better skill alignment leads to stronger project performance. This creates a continuous improvement cycle for the academic project recommendation process.

III. LITERATURE SURVEY

A. Stable Matching for Student Projects

Early academic project allocation studies considered the Student-Project Allocation problem as a matching problem involving student preferences, lecturer preferences, and project capacities. Stable matching techniques can generate assignments that respect specified constraints. Their major contribution is the formalization of project allocation as an algorithmic problem.

These approaches are valuable when the main objective is fair and stable assignment. Their limitation for a skill-based recommendation application is that detailed technical skills may not be modeled explicitly. They therefore provide a foundation but do not completely solve personalized project recommendation.

B. Fair and Preference-Based Allocation

Optimization-based studies have considered fairness, collective satisfaction, and preference ranking. Mixed Integer Linear Programming and related optimization methods can search for high-quality allocations under several constraints. Such work demonstrates that project selection can involve multiple objectives rather than simply filling available project seats.

C. Algorithmic and Approximation Approaches

Research has also investigated the computational complexity of student-project allocation and compared exact optimization with approximation algorithms. These studies show that allocation becomes more difficult as the number of students, projects, preferences, and capacity constraints increases.

D. Skill- and Interest-Based Recommendation

A more personalized direction is to consider student interests and technical skills together. This is directly relevant to a Student Skill-Based Project Recommendation System. A project involving machine learning, for example, may receive a higher score for a student with Python, statistics, and machine-learning knowledge than for a student whose main strengths are networking and web design.

E. Intelligent Project Matching

Recent work has explored intelligent and dynamic matching using student profiles, interests, skill ratings, and project metadata. Such approaches indicate a movement from static preference-based allocation toward adaptive and personalized project recommendation.

F. Recommendation Systems in Education

Recommendation systems are increasingly useful in educational applications because large numbers of learning resources, courses, projects, and opportunities can make manual selection difficult. A project recommendation system can reduce this search burden by ranking alternatives according to individual characteristics.

G. Data Quality and Explainability

A recommendation is only as useful as the information used to create it. Student skill profiles need to be accurate and regularly updated. Project descriptions should also clearly state the skills and technologies required. In addition, a practical academic system should explain why a project was recommended, for example by showing matched skills and interests.

Additional Discussion

An important advantage of skill-based recommendation is that it can balance technical readiness with student interest. A high skill match without interest may reduce motivation, while a high interest match without adequate technical preparation may increase project difficulty. Combining both factors provides a more balanced recommendation. The system can also consider project difficulty so that students receive projects that are appropriate to their current level while still offering opportunities for learning.

Further Discussion

Project information also needs to be maintained carefully. Required skills, preferred technologies, domain, difficulty and expected outcomes should be clearly specified by coordinators or faculty. A well-structured project database makes the matching process more consistent and reduces incorrect recommendations.

The platform can also support administrators through analytics. Aggregated information can show common student skill gaps, popular project domains, technologies frequently requested by students and projects that receive high evaluation scores. Such information can support better planning of future project offerings.

The system can make project selection more systematic by presenting a ranked list rather than forcing students to search through every available project. Each recommendation can include the matched skills, interest area, difficulty level and the reason for its ranking. This improves clarity and gives students a practical basis for discussion with faculty.

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TABLE I — COMPARATIVE ANALYSIS OF REPRESENTATIVE APPROACHES

Ref.	Focus Area	Methodology / Technology	Key Findings	Limitations
[1]	Student-project allocation	Stable matching	Considers preferences and project capacity	Limited technical-skill detail
[2]	Preference-based allocation	Preference ranking / matching	Supports structured allocation	Preferences may not reflect capability
[3]	Fair allocation	MILP / optimization	Improves fairness and satisfaction	Can be computationally complex
[4]	Algorithmic allocation	Integer programming / approximation	Compares exact and approximate methods	Resource needs for large problems
[5]	Skill-based recommendation	Profile and skill matching	Improves personalization and suitability	Needs accurate skill information
[6]	Intelligent matching	ML / reinforcement learning	Uses profiles, interests and project data	Needs sufficient quality data
[7]	Web-based management	Full-stack web application	Centralizes project information	Integration increases system complexity
[8]	Proposed system	Skill matching + recommendation	Ranks projects for each student	Requires implementation and evaluation

IV. LIMITATIONS OF EXISTING APPROACHES

The literature indicates several limitations. First, many allocation methods emphasize preferences and capacity rather than detailed technical skills. Second, optimization models can become complex as the number of students and projects grows. Third, intelligent recommendation depends strongly on the availability and quality of profile data. Fourth, many systems focus on the allocation stage without connecting recommendation with project progress, faculty feedback, and evaluation.

Another important limitation is explainability. Students and faculty may prefer recommendations that show the reasons for a match rather than simply displaying a numerical score. A practical system should therefore provide understandable evidence such as matched skills, interest overlap, difficulty level, and missing skills.

These limitations motivate a recommendation platform that is simple to use, skill-oriented, transparent, and capable of integrating academic guidance.

Additional Discussion

The comparative analysis shows that different research directions solve different parts of the project-selection problem. Stable matching provides allocation stability, optimization improves fairness, and intelligent recommendation provides personalization. A practical academic system can combine these ideas rather than depending on only one technique. The proposed system therefore treats recommendation as the central activity and leaves final approval to academic staff, creating a balance between automation and human judgement.

Further Discussion

A hybrid approach can combine technical skill matching with interest matching and academic guidance. Technical skills represent readiness, interests represent motivation, and faculty review represents academic judgement. Combining these factors can provide a balanced approach that is practical for a college environment.

Security and role-based access are important because student academic information must be visible only to authorized users. Students can manage their own profiles, faculty can view assigned students, and coordinators can maintain project information. Authentication and access control should therefore be included in the implementation.

Student profiles should be kept current because technical abilities change throughout a degree program. New programming languages, tools, certifications, internships, laboratory work and completed projects can be added to the profile. Updating these details allows the recommendation engine to generate more relevant project suggestions.

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V. RESEARCH GAPS

Based on the reviewed approaches, the following research gaps are identified:

1. Limited use of detailed student technical skills in academic project selection. Many approaches give greater importance to preferences, which may not fully represent technical readiness.
2. Lack of personalized project rankings. A single list of projects may not be suitable for students with different skills, interests, and career goals.
3. Limited integration of student interests and technical skills. A project can be technically suitable but uninteresting, or highly interesting but beyond the student's current capabilities.
4. Insufficient explanation of recommendations. Students and faculty need to understand why a project is ranked highly.
5. Limited connection between recommendation and project lifecycle activities such as progress tracking, evaluation, deadlines, and feedback.
6. Limited use of feedback to improve future recommendations. Project performance data can potentially be used to refine matching rules or machine-learning models.

VI. PROPOSED SYSTEM

The proposed Student Skill-Based Project Recommendation System consists of a student profile module, project database, skill-matching engine, recommendation interface, faculty review module, and administrative dashboard.

Student Profile Module: Students enter their programming languages, technical skills, interests, academic background, preferred domains, and career interests.

Project Database: Each project stores project title, domain, required skills, technologies, difficulty level, project description, and expected learning outcomes.

Recommendation Engine: The engine compares student skills and interests with project requirements and generates a suitability score. Projects can then be ranked from highest to lowest suitability.

Faculty Review: Faculty members can inspect recommendations, verify technical feasibility, and guide the student before final project confirmation.

Dashboard and Reports: The system can show recommended projects, matched skills, missing skills, project status, feedback, and evaluation information.

Additional Discussion

The quality of the recommendation can be improved through structured profile collection. Instead of asking students only to select a skill name, the system can allow them to indicate a skill level such as beginner, intermediate or advanced. Evidence such as completed courses, certifications, previous projects and assessment performance can also strengthen the profile. Project records should similarly indicate which skills are essential and which are optional. This structured information can make the matching score more meaningful.

Further Discussion

The first implementation can use a simple weighted score because it is transparent and easy to explain. Later, historical project data can be used to evaluate machine-learning methods. Any advanced model should be compared with the simple baseline using measurable evaluation parameters such as recommendation relevance, student satisfaction and project completion.

In future, the system can be evaluated using real student and project data. The results can be used to improve recommendation rules, discover skill gaps, and understand whether better skill alignment leads to stronger project performance. This creates a continuous improvement cycle for the academic project recommendation process.

Project information also needs to be maintained carefully. Required skills, preferred technologies, domain, difficulty and expected outcomes should be clearly specified by coordinators or faculty. A well-structured project database makes the matching process more consistent and reduces incorrect recommendations.

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VII. RECOMMENDATION PROCESS

The recommendation process can be implemented as a sequence of simple steps:

Step 1 — Collect student information. The system records the student's interests, technical skills, programming languages, academic background, and preferred domain.

Step 2 — Create project profiles. Every project is described using required skills, technologies, domain, difficulty, and expected outcomes.

Step 3 — Match skills. The system compares the skills possessed by the student with the skills required by each project.

Step 4 — Calculate suitability. A simple score can combine skill matching and interest matching. For example, the skill-match percentage can be combined with an interest-match score to produce a final suitability score.

Step 5 — Rank projects. Projects with higher suitability scores are displayed first. The system can also indicate which required skills are already available and which skills may need improvement.

Step 6 — Faculty verification. A faculty member reviews the recommended projects and provides guidance before the student selects a final project.

Step 7 — Feedback and evaluation. After completion, project performance and feedback can be stored for analysis and possible improvement of future recommendations.

VIII. POSSIBLE SYSTEM ARCHITECTURE

A full-stack web application can be used to implement the system. The frontend provides student and faculty dashboards. The backend exposes REST APIs for user authentication, profile management, project management, recommendation, and feedback. A relational database stores students, skills, projects, recommendations, and evaluation data.

The logical flow is: Student Profile → Skill & Interest Extraction → Project Requirement Matching → Recommendation Score → Ranked Projects → Faculty Review → Project Selection → Monitoring & Evaluation.

Role-based access control can be applied so that students view their own profile and recommendations, faculty members review assigned students, and administrators maintain project and user data.

A simple implementation can first use rule-based matching. For example, if a project requires five skills and the student possesses four of them, the skill-match component is 80%. More advanced machine-learning models can be introduced later when historical project data become available.

Additional Discussion

For implementation, the first version can use a rule-based matching engine because it is simple to understand, easy to test and suitable when historical data are limited. Each required skill can be assigned a weight according to importance. Interest overlap and difficulty can be added as separate factors. After sufficient institutional data are collected, machine-learning techniques can be evaluated against the rule-based baseline using recommendation accuracy, satisfaction and project outcomes.

Further Discussion

The platform can also support administrators through analytics. Aggregated information can show common student skill gaps, popular project domains, technologies frequently requested by students and projects that receive high evaluation scores. Such information can support better planning of future project offerings.

The system can make project selection more systematic by presenting a ranked list rather than forcing students to search through every available project. Each recommendation can include the matched skills, interest area, difficulty level and the reason for its ranking. This improves clarity and gives students a practical basis for discussion with faculty.

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IX. FUTURE WORK

The proposed system can be extended in several directions. First, machine-learning models can be trained on historical project choices and outcomes. Data such as selected project, student skills, project completion, evaluation marks, and student feedback may help predict suitable projects more accurately.

Second, the system can include skill-gap analysis. When a student receives a recommendation for a project that requires skills not yet present in the profile, the system can identify those gaps and suggest learning resources or prerequisite courses.

Third, dynamic recommendations can be introduced. Student profiles change as students learn new technologies and complete additional coursework. The recommendation list can therefore be refreshed whenever the profile is updated.

Fourth, the platform can support project teams. Team-level recommendation can consider the combined skills of multiple students and identify projects that match the team's capabilities.

Fifth, explainable recommendation can be improved. Each recommended project can display the percentage of matched skills, relevant interests, project difficulty, and missing skills. This can make the result easier for students and faculty to understand.

Sixth, analytics can be added for coordinators and HODs. Reports may show popular technologies, common student skill gaps, project completion rates, average evaluation scores, and areas in which more project options are required.

X. EXPECTED BENEFITS

The system is expected to reduce the time students spend searching for suitable projects and increase the alignment between student capabilities and project requirements. It can also provide faculty members with structured information for guiding students. From an institutional perspective, centralized data can improve project allocation, monitoring, evaluation, and reporting.

The main expected benefits are improved project suitability, personalized recommendations, better transparency, reduced manual effort, easier faculty guidance, identification of student skill gaps, and a foundation for future intelligent project allocation.

XI. EVALUATION PARAMETERS

The system can be evaluated using recommendation accuracy, matching quality, student satisfaction, faculty satisfaction, project completion rate, usability, response time, scalability, and the relationship between recommendation suitability and project performance. These measures can be compared before and after implementation.

Additional Discussion

Another useful extension is team-based project recommendation. In group projects, the system can combine the skills of several students and compare the overall team profile with project requirements. It can identify whether the team has adequate coverage for programming, database, documentation, testing and presentation activities. This can help coordinators form balanced teams and reduce the risk that one student is responsible for most of the technical work.

Further Discussion

Security and role-based access are important because student academic information must be visible only to authorized users. Students can manage their own profiles, faculty can view assigned students, and coordinators can maintain project information. Authentication and access control should therefore be included in the implementation.

Student profiles should be kept current because technical abilities change throughout a degree program. New programming languages, tools, certifications, internships, laboratory work and completed projects can be added to the profile. Updating these details allows the recommendation engine to generate more relevant project suggestions.

The first implementation can use a simple weighted score because it is transparent and easy to explain. Later, historical project data can be used to evaluate machine-learning methods. Any advanced model should be compared with the simple baseline using measurable evaluation parameters such as recommendation relevance, student satisfaction and project completion.

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XII. CONCLUSION

The Student Skill-Based Project Recommendation System provides a personalized approach to academic project selection. Instead of depending only on student preference or manual allocation, the system compares technical skills and interests with project requirements and produces ranked project recommendations. This can help students choose projects that are both relevant and achievable.

The literature indicates that stable matching, preference-based allocation, optimization, and intelligent matching each contribute useful ideas, but there remains value in a practical system centered on student skills, personalized recommendation, faculty guidance, and explainable results. The proposed architecture provides a foundation for such a platform and can later be extended with machine learning, skill-gap analysis, team recommendation, monitoring, evaluation, and analytics.

Overall, a Student Skill-Based Project Recommendation System can improve transparency and efficiency in academic project selection while helping students work on projects that better match their technical abilities and interests.

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Additional Discussion

A successful implementation should finally be evaluated with real users. Students can rate whether the recommendations are relevant, understandable and useful. Faculty members can evaluate whether the recommendations reduce manual effort and support project guidance. The institution can measure completion rates, project evaluation marks and time required for allocation. These results can be used to refine the system and establish whether skill-based recommendation provides measurable benefits in an academic environment.

Further Discussion

In future, the system can be evaluated using real student and project data. The results can be used to improve recommendation rules, discover skill gaps, and understand whether better skill alignment leads to stronger project performance. This creates a continuous improvement cycle for the academic project recommendation process.

Project information also needs to be maintained carefully. Required skills, preferred technologies, domain, difficulty and expected outcomes should be clearly specified by coordinators or faculty. A well-structured project database makes the matching process more consistent and reduces incorrect recommendations.

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